Politecnico di Milano

Dipartimento di Elettronica, Informazione e Bioingegneria

prof. Fabrizio Ferrandi

Advanced Algorithms and Parallel Programming 1st part–July 4th 2025

Polimi ID _____ _____

Surname _____ _____ **Name** _____

- This is a closed-book examination. You cannot use computers, phones, or laptops during the exam.
- Paper will be provided, but you should bring and use writing instruments that yield marks dark enough to be read easily. Erasable pens can be used.
- Total available time: 1h:00m.

Exercise 1 (4 points) _____

Exercise 2 (4 points) _____

Exercise 3 (4 points) _____

Exercise 4 (4 points) _____

Exercise n. 1

Answer the following questions about parallel application modeling and ensure you provide a brief explanation. Remember, without a reason, the answer will be considered invalid.

A. What does it imply that each memory cell can hold an integer of unbounded size in a PRAM? (1)

B. Given an algorithm with $T_1(n)=n^3$ and $T_p(n)=n^3/p$, analyze the speedup and efficiency. What can you conclude? (1)

C. Why is the prefix sum algorithm on PRAM more efficient than the basic sum algorithm, and what is the trade-off? (1)

D. How does Gustafson's Law support the concept of weak scaling, and what is its implication for massively parallel systems? (1)

Exercise n. 2

Answer the following questions about advanced algorithms and briefly explain (without an explanation, the answer will be considered invalid)

A. In class, we studied an algorithm that shows the main hallmarks of dynamic programming. Describe the algorithm and its complexity, and why it is classified as a good dynamic programming example. (2)

B. Please list the main differences, including the efficiency, between the Las Vegas and Monte Carlo algorithms. (2)

Exercise n. 3

Answer the following questions about advanced algorithms and briefly explain (without an explanation, the answer will be considered invalid)

A. Describe an algorithm seen during the course lectures where the potential method has been used. (2)

B. Describe how the union and the split work in a treap and discuss its complexity. (2)

Exercise n. 4

Answer the following questions about the algorithm analysis and briefly explain (without an explanation, the answer will be considered invalid)

A. Please describe the partitioning algorithm used by the quick sort and its key properties. How is it randomized? (1) **This question can be skipped if you got 1 or 3 points in the first challenge.**

B. What are the differences between the randomized and deterministic algorithms for solving the Select-ith problem? (2) **This question can be skipped if you got 2 or 3 points in the first challenge.**

C. What are the characteristics of an algorithm that require the amortized analysis methods? (1)